DB LAB 08 SUBMISSION

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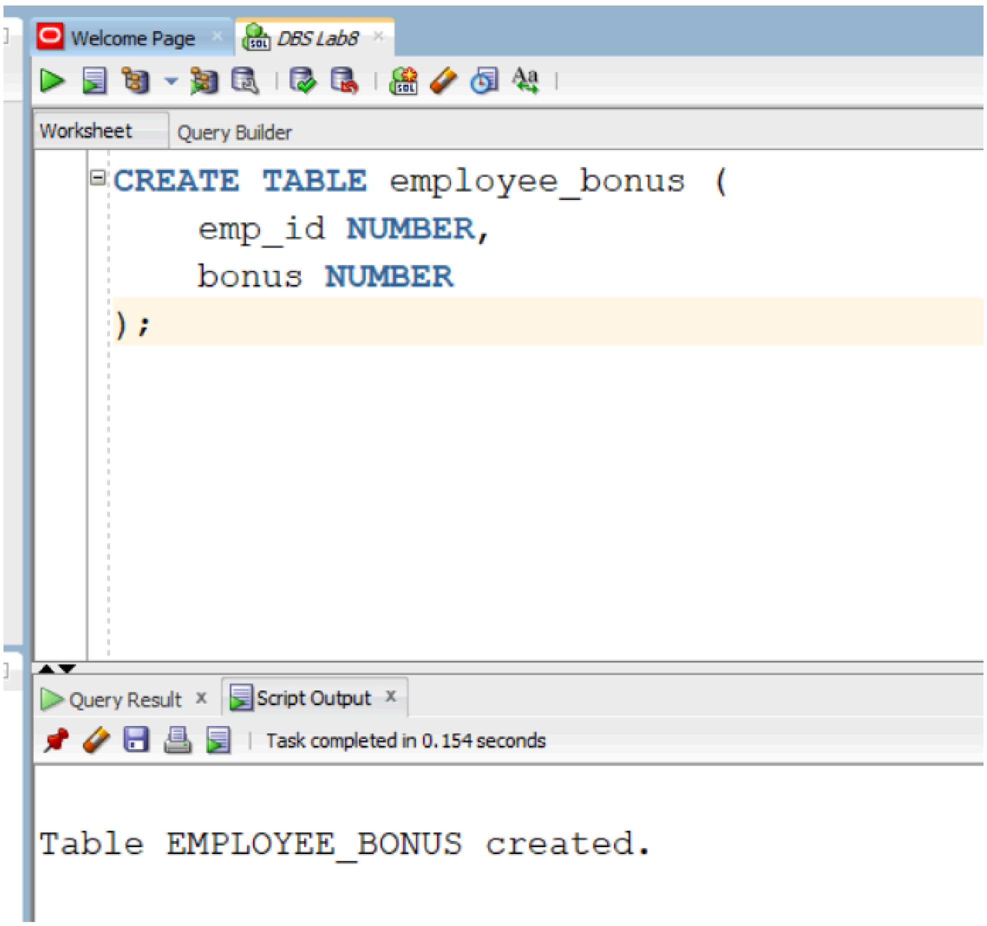
Task 01:



```sql
CREATE OR REPLACE TRIGGER trg_emp_bonus
AFTER INSERT ON employees
FOR EACH ROW
ENABLE
BEGIN
INSERT INTO employee_bonus (emp_id, bonus)
VALUES (:NEW.employee_id, :NEW.salary * 0.10);
END;
/
```

Script Output ×

Task completed in 0.075 seconds

```
Trigger TRG_EMP_BONUS compiled
```

Task 2:

```sql
CREATE OR REPLACE TRIGGER trg_salary_check
BEFORE UPDATE ON employees
FOR EACH ROW
ENABLE
BEGIN
IF :NEW.salary > 10000 THEN
RAISE_APPLICATION_ERROR(-20001, 'Salary cannot exceed 10000');
END IF;
END;
/
```

Script Output ×

Task completed in 0.075 seconds

```
Trigger TRG_SALARY_CHECK compiled
```

Task 3:

```sql
CREATE TABLE Deleted_Employees_Log (
    emp_id NUMBER,
    emp_name VARCHAR2(30),
    salary NUMBER,
    deleted_date DATE
);
```

Script Output

Task completed in 0.055 seconds

```
Trigger TRG_SALARY_CHECK compiled


Table DELETED_EMPLOYEES_LOG created.
```

```sql
CREATE OR REPLACE TRIGGER trg_delete_emp
BEFORE DELETE ON employees
FOR EACH ROW
ENABLE
BEGIN
INSERT INTO Deleted_Employees_Log
VALUES (:OLD.employee_id, :OLD.first_name, :OLD.salary, SYSDATE);
END;
/
```

Script Output

Task completed in 0.071 seconds

```
Trigger TRG_DELETE_EMP compiled
```

Task 4:

```sql
CREATE TABLE salary_log (
    emp_id NUMBER,
    old_salary NUMBER,
    new_salary NUMBER,
    change_date DATE
);
```

Script Output ×

Task completed in 0.051 seconds

```
Table SALARY_LOG created.
```

```sql
CREATE OR REPLACE TRIGGER trg_salary_log
BEFORE UPDATE ON employees
FOR EACH ROW
ENABLE
BEGIN
INSERT INTO salary_log
VALUES (:OLD.employee_id, :OLD.salary, :NEW.salary, SYSDATE);
END;
/
```

Script Output ×

Task completed in 0.073 seconds

```
Trigger TRG_SALARY_LOG compiled
```

DDL Trigger Tasks:

Task 1:

```sql
--DDL Trigger Tasks:
--1:
CREATE TABLE Audit_Log (
    object_name VARCHAR2(30),
    created_by VARCHAR2(30),
    created_date DATE
);
```

Script Output

Task completed in 0.05 seconds

```
Table AUDIT_LOG created.
```

```sql
CREATE OR REPLACE TRIGGER trg_create_table
AFTER CREATE ON SCHEMA
BEGIN
INSERT INTO Audit_Log
VALUES (ora_dict_obj_name,
        sys_context('USERENV','CURRENT_USER'),
        SYSDATE);
END;
/
```

Script Output × | Task completed in 0.086 seconds

```
Trigger TRG_CREATE_TABLE compiled
```

Task 02:

```sql
--2:
CREATE OR REPLACE TRIGGER trg_no_alter_after_hours
BEFORE ALTER ON SCHEMA
BEGIN
IF TO_CHAR(SYSDATE,'HH24') NOT BETWEEN '08' AND '18' THEN
IF ora_dict_obj_name = 'EMPLOYEES' THEN
RAISE_APPLICATION_ERROR(-20002, 'Cannot ALTER Employees table after business hours');
END IF;
END IF;
END;
/
```

Script Output × | Task completed in 0.064 seconds

```
Trigger TRG_NO_ALTER_AFTER_HOURS compiled
```

Task 03:

```sql
--3:
CREATE TABLE Drop_Log (
    object_name VARCHAR2(30),
    dropped_by VARCHAR2(30),
    drop_time DATE
);
```

Script Output ×

Task completed in 0.046 seconds

```
Table DROP_LOG created.
```

```sql
CREATE OR REPLACE TRIGGER trg_drop_log
AFTER DROP ON SCHEMA
BEGIN
INSERT INTO Drop_Log
VALUES (ora_dict_obj_name,
        sys_context('USERENV','CURRENT_USER'),
        SYSDATE);
END;
/
```

Script Output ×

Task completed in 0.075 seconds

```
Trigger TRG_DROP_LOG compiled
```

Task 04:

```sql
--4:
CREATE OR REPLACE TRIGGER trg_prevent_drop_audit
BEFORE DROP ON SCHEMA
BEGIN
IF ora_dict_obj_name = 'AUDIT_LOG' THEN
RAISE_APPLICATION_ERROR(-20003, 'Audit_Log table cannot be dropped');
END IF;
END;
/
```

Script Output

Task completed in 0.061 seconds

```
Trigger TRG_PREVENT_DROP_AUDIT compiled
```

System/Database Trigger Tasks:

Task 01:

```sql
--System/Databse Trigger Tasks:
--1:
CREATE TABLE System_Logs (
    event_type VARCHAR2(30),
    log_time DATE
);
```

Script Output ×

Task completed in 0.046 seconds

```
Table SYSTEM_LOGS created.
```

```sql
CREATE OR REPLACE TRIGGER trg_startup_log
AFTER STARTUP ON SCHEMA
BEGIN
INSERT INTO System_Logs
VALUES (ora_sysevent, SYSDATE);
END;
/
```

Script Output ×

Task completed in 0.061 seconds

```
Trigger TRG_STARTUP_LOG compiled
```

Task 02:

```
--2:
CREATE TABLE Failed_Logins (
    username VARCHAR2(30),
    attempt_time DATE
);
```

Script Output

Task completed in 0.049 seconds

```
Table FAILED_LOGINS created.
```

```
CREATE OR REPLACE TRIGGER trg_failed_login
AFTER LOGON ON SCHEMA
BEGIN
NULL;
END;
/
```

Script Output

Task completed in 0.049 seconds

```
Trigger TRG_FAILED_LOGIN compiled
```

Task 03:

```sql
--3:
CREATE TABLE User_Activity_Log (
    username VARCHAR2(30),
    logoff_time DATE
);
```

Script Output

Task completed in 0.048 seconds

```
Table USER_ACTIVITY_LOG created.
```

```sql
CREATE OR REPLACE TRIGGER trg_logoff
BEFORE LOGOFF ON SCHEMA
BEGIN
INSERT INTO User_Activity_Log
VALUES (USER, SYSDATE);
END;
/
```

Script Output

Task completed in 0.059 seconds

```
Trigger TRG_LOGOFF compiled
```

Instead of Trigger Tasks:

Task 01:

```
--Instead of Trigger Tasks:
--1:
CREATE VIEW emp_dept_view1 AS
SELECT e.employee_id, e.first_name, e.salary, d.department_id, d.department_name
FROM employees e, departments d;
```

Script Output

Task completed in 0.064 seconds

```
View EMP_DEPT_VIEW1 created.
```

```
CREATE OR REPLACE TRIGGER trg_io_insert1
INSTEAD OF INSERT ON emp_dept_view1
FOR EACH ROW
BEGIN
INSERT INTO employees (employee_id, first_name, salary)
VALUES (:NEW.employee_id, :NEW.first_name, :NEW.salary);
INSERT INTO Departments (department_id, department_name)
VALUES (:NEW.department_id, :NEW.department_name);
END;
/
```

Script Output

Task completed in 0.087 seconds

```
Trigger TRG_IO_INSERT1 compiled
```

Task 02:

```
--2:
CREATE VIEW emp_salary_view AS
SELECT employee_id, salary FROM employees;
```

Script Output

Task completed in 0.048 seconds

```
View EMP_SALARY_VIEW created.
```

```
CREATE OR REPLACE TRIGGER trg_io_update_salary
INSTEAD OF UPDATE ON emp_salary_view
FOR EACH ROW
BEGIN
IF :NEW.salary < (:OLD.salary * 0.80) THEN
RAISE_APPLICATION_ERROR(-20004, 'Salary reduction exceeds 20%');
ELSE
UPDATE employees
SET salary = :NEW.salary
WHERE employee_id = :OLD.employee_id;
END IF;
END;
/
```

Script Output

Task completed in 0.066 seconds

```
Trigger TRG_IO_UPDATE_SALARY compiled
```